

COMP5048 Assignment 1B: Individual Work**By:****Prof. Seokhee Hong****Dr. Amyra Meidiana****COMMONWEALTH OF AUSTRALIA****Copyright Regulations 1969****WARNING**

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COMP5048 Visual Analytics 2025 S1 Assignment 1B

Deadline: April 10 (Week 7) Thursday 23:59pm (pdf and video on Canvas)

Construct good visualisations of TWO data (one graph from each category):

- You can use any suitable visualisation method chosen from the tools assigned to each category.
- Create visualisations for each data according to the instructions given.
- In your report, explain your justification for your selected visualisation and analysis methods, then evaluate and compare the pros and cons of your visualisations.

Category A:

For this category, visualise the data using any tool from the following: Tulip, yEd, Graphviz

A1: Greek Gods

A2: Software Dependency

For the selected graph:

- Visual Analysis 1:
 - Analyse the graph to identify the important entities.
 - Visualise the graph using a suitable layout, displaying the analysis result.
- Visual Analysis 2: Using a different analysis method and different suitable layout:
 - Analyse the graph to identify the important entities.
 - Visualise the graph using a suitable layout, displaying the analysis result.

Category B:

For this category, visualise the data using any tool from the following: NetworkX, Gephi, D3

B1: Workplace Contacts (Second Year)

B2: Primary School Contacts

For the selected graph:

- Visual Analysis 1:
 - Analyse the graph to identify the cohesive groups in each time step.
 - Visualise all time steps of the graph displaying the analysis result.
- Visual Analysis 2: Using different analysis method and different visualisation method:
 - Analyse the graph to identify the cohesive groups in each time step.
 - Visualise all time steps of the graph displaying the analysis result.
- Visual Analysis 3:
 - Compute Animations of Visual Analysis 1 and Visual Analysis 2.
 - Include a video of your animation in your submission, with screenshots in the report.

Submission: Minimum 10 - page report

- **Category A:**
 - 1st page: Visual Analysis 1
 - 2nd page: Discussion of Visual Analysis 1
 - 3rd page: Visual Analysis 2
 - 4th page: Discussion of Visual Analysis 2
- **Category B:**
 - 5th page: Visual Analysis 1
 - 6th page: Discussion of Visual Analysis 1
 - 7th page: Visual Analysis 2
 - 8th page: Discussion of Visual Analysis 2
 - 9th page: Visual Analysis 3
 - 10th page: Discussion of Visual Analysis 3
- Write your Discussion of each Visual Analysis with the following headings:
 - **Design:** tools and layouts with justification (design choice)
 - **Evaluation:** explain the findings you can see from each visualisation
 - **Pros and Cons**
 - **Comparison:** compare different Visual Analysis of the same data (i.e., one comparison in 4th page for Category A and one comparison in 10th page for Category B)
- Acknowledge all your sources in **References**
- You must acknowledge any **usage of AI**

Appendix (not included in page count):

- Provide code, if applicable

Marking Rubric: (total 12 marks)

- Category A (total 3 marks): per Visual Analysis (1.5 marks)
 - Quality of Visualisation: 0.8 marks
 - Quality of Discussion: 0.7 marks
- Category B (total 9 marks): per Visual Analysis (3 marks)
 - Quality of Visualisation: 1.5 marks
 - Quality of Discussion: 1.5 marks